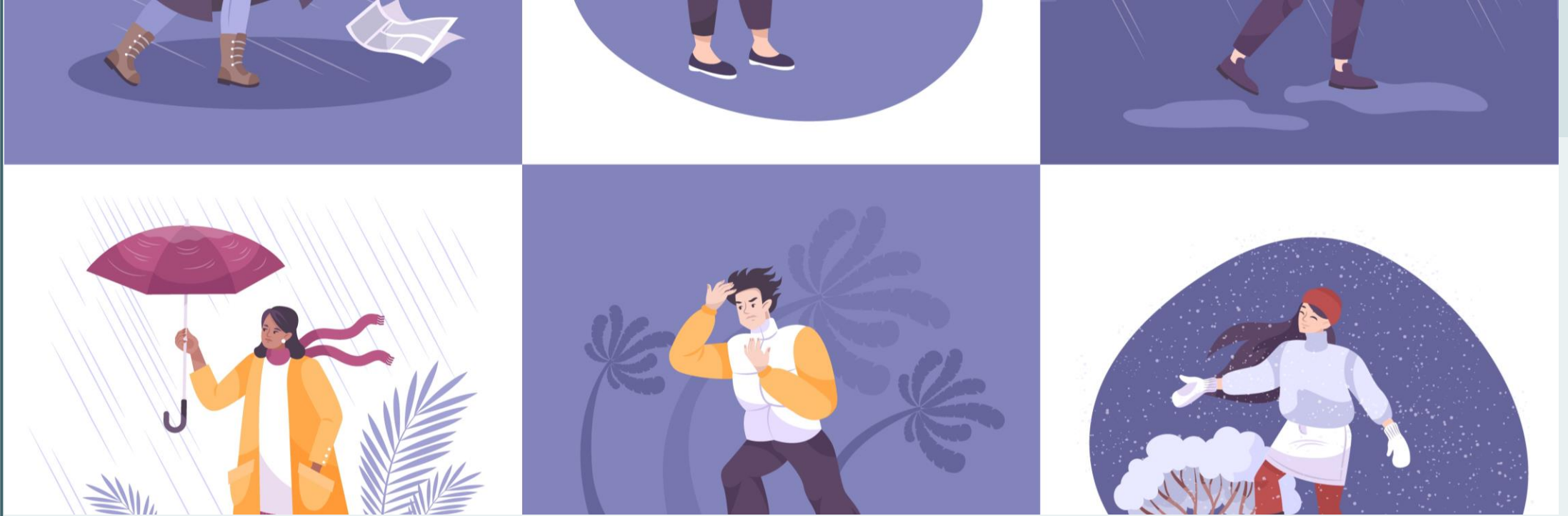




Title: Weather Dashboard Web Application

Course: DLBCSPJWD01 – Project Java and Web Development

Student: Muhammad Saad Bin Khurram

Matriculation Number:10220341

Purpose of the Application



Real-time weather updates for any city using the OpenWeatherMap API.

Quick city search through a simple and intuitive search bar.

Save favorite cities for faster access to frequent locations.

Responsive design for seamless use on desktop, tablet, or mobile.

Useful for students, travelers, and commuters who plan their day around weather conditions.

Target Users & Benefits

Target Users:

- Students & employees (to plan daily routines).
- Travelers (to check destination weather).
- General users who want quick updates.

Benefits:

- Saves time by providing instant forecasts.
- Favorites list gives quick access to frequently checked locations.
- Responsive design ensures usability on desktop, tablet, and mobile.



Technical Design Choices

- Frontend:** HTML, CSS, JavaScript → Responsive UI.
- Backend:** Node.js (Express) → Handles API requests.
- Database:** PostgreSQL → Stores favorite cities.
- API Integration:** OpenWeatherMap (free API).
- Dynamic Aspects:**
 1. Search city → fetch & display weather.
 2. Save/remove cities in “Favorites” list.



Summary & Next Steps

The Weather Dashboard is a simple and user-friendly web application that helps people check real-time weather updates for any city around the world. It combines a clean, responsive front-end design with a reliable Node.js back-end connected to the OpenWeatherMap API.

With just a quick search, users can view essential weather details like temperature, humidity, and wind speed, along with easy-to-understand weather icons. The app also allows users to save their favorite cities in a personal list, making it effortless to check the weather in those places again later.

Overall, the project focuses on making weather information fast, clear, and accessible from any device whether you're at home, at work, or on the go.

In the next phase, I will begin implementing the core features and connecting the app to the weather API